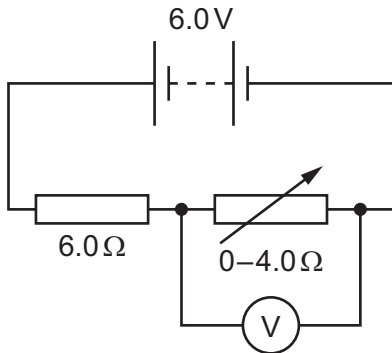


- 37** A battery of electromotive force (e.m.f.)  $6.0\text{ V}$  and negligible internal resistance is connected in series with a resistor of resistance  $6.0\ \Omega$  and a variable resistor of resistance from zero to  $4.0\ \Omega$ . A voltmeter is connected across the variable resistor. The resistance of the variable resistor is changed.



What is the range of the voltmeter reading?

- A**  $0\text{ V}-2.4\text{ V}$       **B**  $0\text{ V}-3.6\text{ V}$       **C**  $2.4\text{ V}-6.0\text{ V}$       **D**  $3.6\text{ V}-6.0\text{ V}$